		$\frac{\quad}{(1)}$: $\tan\theta = \frac{\quad}{50} \rightarrow \theta = 11$
3	**B**	By relative speed of approach = relative speed of separation, $2.0 - 0 = v - (-0.50)$ $v = 1.5 \text{ m s}^{-1}$ By conservation of linear momentum, $m_P(2.0) = m_P(-0.50) + m_Q v$ $2.5 m_P = m_Q v$ $\frac{m_P}{m_Q} = \frac{1.5}{2.5}$ $= \frac{3}{5}$

No.	Answer	Solution
4	B	$U = T + mg$